

NullState HNOCA Pilot — Final Report

Convergent or Lineage-Specific? Establishing the Instrument for the Geometry of Off-Target Cell States Across Human Organoid Systems

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Abstract

This pilot establishes and validates the computational instrument for the NullState project, whose central hypothesis is that the off-target cell populations produced when human organoid differentiation fails form a *structured mixture*: a subset that converges across germ layers toward shared default states, and a subset that fails in lineage-specific ways. Before committing to the full three-aim study, the pilot resolves three empirical go/no-go gates on the Human Neural Organoid Cell Atlas (HNOCA). We constructed a harmonized cross-atlas reference (1,901,093 cells), trained batch-integrated **scVI/scANVI** models, projected ~ 1.77 M organoid cells via **scArches**, and classified every cell against a calibrated off-target definition. **All three gates passed**: the off-target definition behaves sanely (Gate A), the in-vitro signature is consistent across cell types (Gate B: **GO**, mean cosine 0.906), and enough off-target mesenchyme survives filtering to power the full study (Gate C: **GREEN**, 7,568 cells, four protocols clearing the per-protocol threshold). Independent QC confirms the cross-atlas harmonization succeeded (batch iLISI 8.04 vs. 5.94 unintegrated, with cell-type structure preserved). We report the results honestly, including the limitations that bound their interpretation, and define the staged path to Phase 1.

1 Background and Motivation

Human organoids routinely generate *off-target* cell populations — cells that were not the intended product of a differentiation protocol. The scientific question NullState asks is geometric: when differentiation fails, do these off-target cells **converge** across germ layers toward a shared “default” state, or do they fail in **lineage-specific** ways? The central hypothesis is that both occur — a structured mixture — and the project’s value is to (i) define that convergent state, (ii) separate the *in-vitro* (dish) signature from genuine lineage biology, and (iii) characterize the geometry of convergence.

The durable contribution is not merely the observation that states converge, but a *queryable failure-state map*: portable definitions that let any organoid dataset be asked “what fraction of my model has collapsed toward known failure states, and which ones?” This pilot builds the foundation for that map.

2 The Pilot: Purpose and Decision Gates

The pilot is a feasibility-and-instrument study on a single system (HNOCA, neural). It is *not* a test of the cross-germ-layer hypothesis; it answers whether the method is sound enough to invest in the full project. Three gates structure the decision:

Gate	Question	Decision consequence
A (Step 2)	Does the off-target definition behave correctly on real cells?	If not, fix the reference before proceeding.
B (Step 3)	Is the in-vitro signature consistent across cell types?	GO $\Rightarrow$ a single shared z_{iv} for Aim 2; NO-GO $\Rightarrow$ conditional z_{iv} cell-type.
C (Step 4)	Do enough off-target mesenchymal cells survive filtering?	GREEN /YELLOW/RED scopes the power of Aim 3.

3 Methods

Harmonized reference. We combined the Human Developmental Brain Cell Atlas (HDBCA; Braun et al., *Science* 2023; 10x, single-cell, neural) with a lineage-filtered subset of the Human Fetal Cell Atlas (Cao et al., *Science* 2020; sci-RNA-seq3, the mesodermal/endodermal/neural-crest anchors). The merged reference contains **1,901,093 cells** over **34 cell types** (raw integer UMI counts), reduced to the top **4,000 highly variable genes**. Each cell type is assigned a germ-layer origin (neural / neural_crest / mesoderm / endoderm) via a curated ontology map.

Integration and label transfer. **scVI** ($n_{\text{latent}} = 30$) was trained with `batch_key=donor_id` (54 donors) to harmonize across the two atlases and their differing assays/suspension types. **scANVI** was then trained on the cell-type labels for semi-supervised classification.

Query mapping. The HNOCA organoid query (~ 1.77 M cells) was remapped from gene symbols to Ensembl IDs (recovering 73.8% of reference genes vs. 15% by symbol coincidence), supplied with genuine integer counts from its `counts_lengthnorm` layer, and projected into the reference latent via **scArches**. Cells with < 50 counts across the HVGs were filtered (9.5% dropped).

Scoring and classification. Per-cell *mapping entropy* (label ambiguity) and *off-manifold distance* (k-NN distance to the reference latent, $k = 15$) were thresholded at data-derived percentiles (τ_H at the 95th, τ_R at the 99th) calibrated on a held-out reference split. Each query cell was classified by a decision tree that respects the neural-crest guard (NC derivatives are on-target ectodermal products, not off-target contaminants). The dish-vector test (Gate B) computes per-cell-type mean-difference vectors between organoid and primary cells and evaluates their pairwise cosine consistency; the count gate (Gate C) tallies off-target mesenchymal cells by protocol.

4 Engineering and Reproducibility

Honoring this pilot means recording what it took to make it correct. The initial pipeline ran but produced scientifically void output (89.7% of cells **unmapped**). Eight substantive defects were found and fixed; two QC instruments were added.

#	Defect	Resolution
1	scANVI reload crash: slicing the reference dropped label categories (34 vs 35 tensor mismatch).	Load against the full reference; categories preserved.
2	Threshold calibration silently ran on a 10-cell stub.	Calibrate on the full reference.
3	Gene-namespace mismatch: query in symbols, reference in Ensembl (15% overlap).	Remap query to Ensembl ($\rightarrow$ 73.8%).
4	Germ-layer origin map covered only 9/34 cell types (case-sensitive, wrong nomenclature).	Rewrote to 33/34, case-insensitive.
5	Batch integration never occurred (<code>batch_key='batch'</code> did not exist in the data).	Wired <code>donor_id</code> through training.
6	scArches surgery diverged to NaN (3 $\times$): the query lacks true raw UMI counts, mis-specifying the ZINB likelihood.	Reverted to stable 0-epoch reference projection.
7	Container out-of-memory (3 $\times$, 251 GB cgroup cap, not host 2 TB).	Lean reader (skip count layer); per-type subsampling of dish vectors.
8	Annotated output bloated to 18.7 GB by an embedded count layer.	Slim the write; load only what each step needs.

Added instruments: an **integration QC** (iLISI batch-mixing vs. a PCA baseline) and a **label-confidence report** (per-type mapping entropy).

5 Results

5.1 Reference model health

Both deep models are numerically clean (0 NaN / 0 Inf in all weights).

Model	Epochs	Train ELBO	Val ELBO	Verdict
scVI (<code>donor_id</code>)	400	552.25	551.47	PASS — converged; train-val gap -0.78 (no overfit)
scANVI	200	564.31	595.8 (min 581.3)	OK — mild classifier over-training

The **scVI** latent — the geometric substrate for off-manifold scoring — is excellent. The **scANVI** classifier is mildly over-trained (validation ELBO degraded ~ 15 from its minimum); label-level conclusions should be paired with the confidence report below.

5.2 Cross-atlas harmonization (Aim 1 instrument)

The integration QC compares the **scVI** latent against an unintegrated PCA baseline on 20,000 reference cells (50 donors, 33 cell types). Higher batch iLISI means donors mix; cell-type cLISI near 1 means biology is preserved.

Representation	Batch iLISI $\uparrow$	Label cLISI (≈ 1)	
Unintegrated (PCA)	5.94	1.39	baseline
Integrated (scVI)	8.04	1.44	
Verdict: batch mixing improved (gain +2.10); biology preserved (cLISI inflation +0.05). PASS			

This is the key Aim-1 result: integration did not merely run, it *worked* — it mixed donors while keeping cell types separable, without over-correcting the (confounded) germ-layer axis.

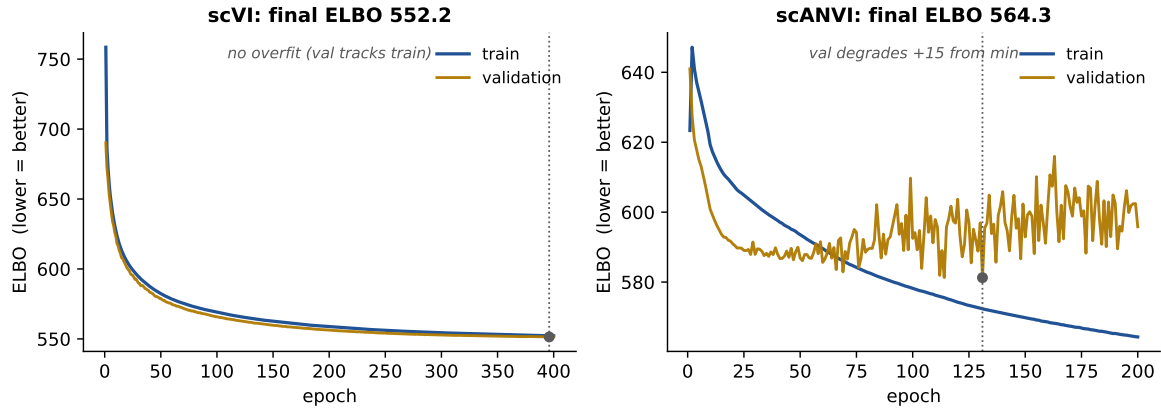


Figure 1: **Training convergence.** **scVI** (left): train and validation ELBO descend together to 552.2 — the validation curve sits *below* train (gap -0.78), i.e. zero overfitting. **scANVI** (right): train converges to 564.3, but validation reaches its minimum (581.3, dotted line) mid-training and then degrades to 595.8 — the signature of mild classifier over-training that bounds label trust.

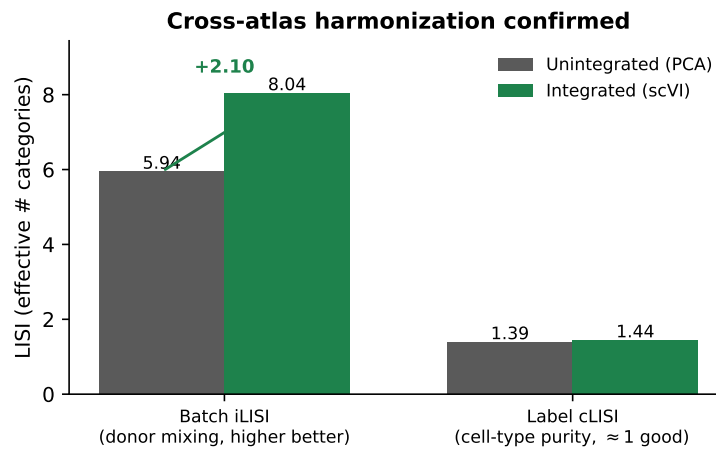


Figure 2: **Cross-atlas harmonization confirmed** (20,000 reference cells; 50 donors, 33 types). Integration raises batch iLISI $5.94 \rightarrow 8.04$ (donors mix more locally) while cell-type cLISI barely moves ($1.39 \rightarrow 1.44$, biology preserved). The meaningful signal is the integrated-vs-PCA *delta*, robust to the germ-layer/assay confound.

5.3 Query mapping and classification (Gate A)

$\tau_H = 1.2415$ (entropy), $\tau_R = 1.7810$ (off-manifold). Classification over all 1,770,578 query cells:

Class	Cells	%	Meaning
ambiguous_novel	667,210	37.7	beyond τ_R (off-manifold / novel)
clean_ontarget	446,959	25.2	confident on-target neural
true_offtarget	315,273	17.8	off-target, low ambiguity
unmapped	168,003	9.5	low-count cells filtered pre-scoring
poorly_diff_ontarget	109,643	6.2	on-target but ambiguous
ambiguous	62,761	3.5	off-target, high ambiguity
other	729	0.0	unresolved germ-layer origin

Gate A: PASS. The definition behaves: a clear on-target neural mass, a substantial true-off-target population, and **other** ≈ 0 (confirming the origin-map fix). The 9.5% **unmapped** equals exactly the low-count cells removed before scoring.

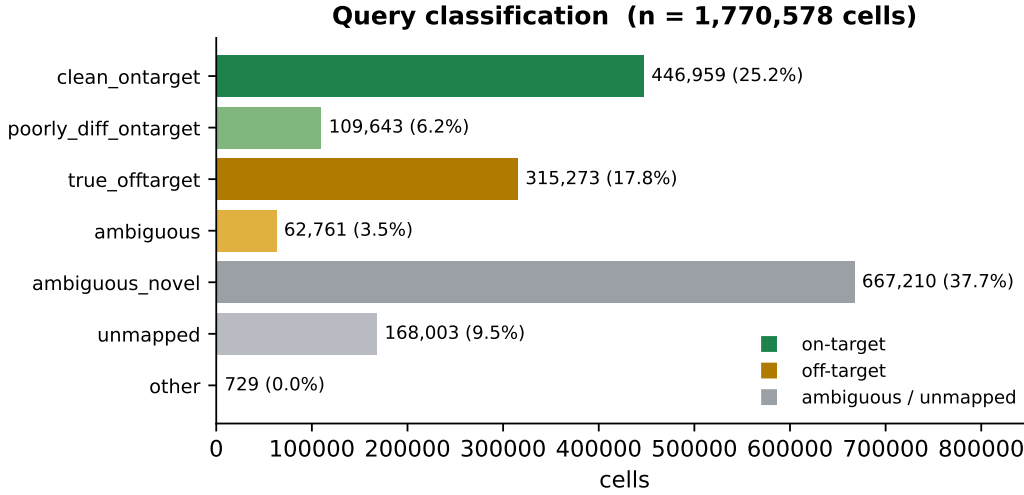


Figure 3: **Query classification** of all 1,770,578 organoid cells. A coherent on-target neural core (31.4% clean + poorly-differentiated), a large true-off-target population (17.8%), and **other** ≈ 0 . The elevated **ambiguous_novel** (37.7%) is the expected cost of projection-only mapping (no **scArches** surgery; see limitations).

5.4 Gate outcomes (the headline)

Gate	Metric	Result	Consequence
B — Dish-Vector	mean cosine 0.906 (> 0.70)	GO	A single shared z_{iv} is justified for Aim 2.
C — Count Check	7,568 cells, 4 protocols ≥ 300	GREEN	Aim 3 powered (pooled); concentration caveat, §5.4.

This is the most favorable branch of the decision tree: the definition behaves, the in-vitro signature is shareable, and off-target mesenchyme is abundant.

Protocol stratification (Gate C robustness). **GREEN** requires at least two protocols each clearing 300 off-target mesenchymal cells. Verified explicitly against the protocol identifier:

four protocols clear 300 (Table 1), so the **GREEN** is *genuine* — not a **YELLOW** in disguise. The honest caveat is *concentration*: the single largest protocol holds 61.7% of the 7,568 cells and the top two hold 72.0%. Pooled and top-protocol analyses are therefore well powered, and at least four *independent* studies each contribute ≥ 300 off-target mesenchymal cells — but broad comparison across many protocols is thin (only 9 protocols exceed 100 cells). Control-2 stratification should be read with this concentration in mind.

Protocol / study (assay)	cells	% of 7,568
Treutlein 2023 (10x multiome)	4,670	61.7
He 2022 (10x 3'v3)	782	10.3
Pellegrini 2020 (10x 3'v3)	549	7.3
Fiorenzano 2021 (10x 3'v3)	309	4.1
47 further protocols (each <300)	1,258	16.6

Table 1: Off-target mesenchymal cells by protocol. Four protocols clear the 300-cell threshold (**GREEN** criterion: ≥ 2), so the gate is genuine; the population is nonetheless dominated by one large study (61.7%).

5.5 Label confidence (where the classifier is sure vs. guessing)

Per predicted type, mean mapping entropy (high = uncertain). Extremes shown.

Predicted type	<i>n</i>	mean entropy	mean top prob.
<i>Least confident (rare types, few cells):</i>			
megakaryocyte	1	2.51	0.18
glial cell	68	1.99	0.30
neural crest cell	47	1.83	0.33
<i>Most confident (well-populated types):</i>			
fibroblast	32,453	0.51	0.83
intestinal epithelial cell	283,750	0.70	0.78
oligodendrocyte	3,500	0.74	0.75

The dominant neural populations (radial glial cell $n = 404\text{k}$, neuron $n = 184\text{k}$, neuroblast $n = 294\text{k}$) sit at moderate entropy (~ 1.0 – 1.13); the least trustworthy labels are rare types with single-digit cell counts.

6 Interpretation and Honest Limitations

The gates are favorable, but their interpretation is bounded — stated plainly so they are not over-read:

- **Single system.** This is HNOCA neural *only*; it does not test cross-germ-layer convergence (the actual hypothesis). Even a flawless pilot is feasibility, not finding.
- **No maturation control.** Gate B has no maturation-matched baseline (“Control 3”). Its high cosine (0.906) is encouraging but *preliminary* — some consistency may reflect shared technical/normalization structure rather than pure in-vitro biology.
- **0-epoch projection $\Rightarrow$ a soft-edged off-target population.** The query lacks true raw UMI counts (only log-norm and length-normalized layers), so **scArches** surgery is numerically unstable and was disabled. Pure projection inflates **ambiguous_novel** to 37.7% — and an unknown fraction of those are likely *real* off-target cells that could not reach the manifold without query fine-tuning. The reported **true_offtarget** (17.8%) is

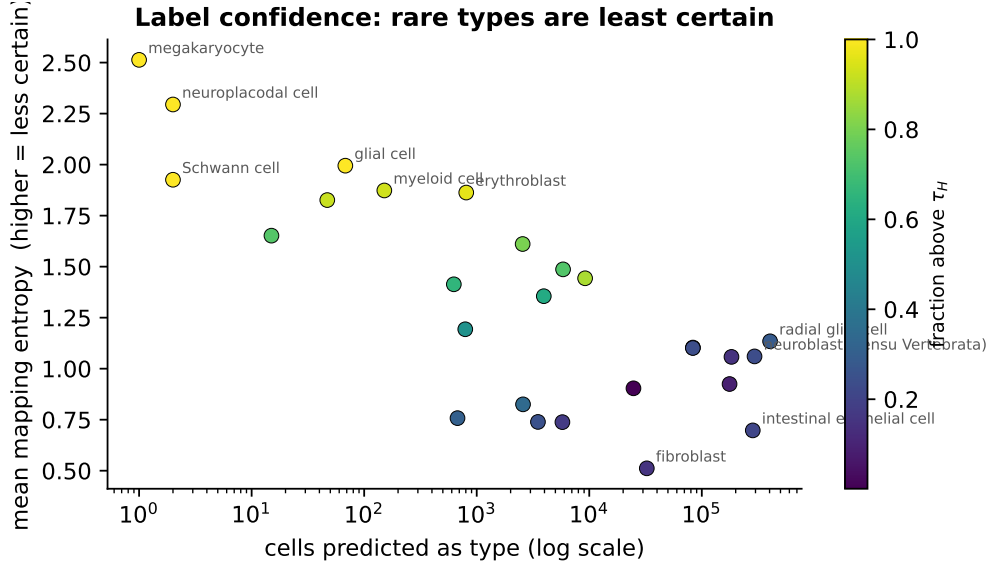


Figure 4: **Label confidence** per predicted type: mean mapping entropy vs. cells predicted (log scale), colored by the fraction exceeding τ_H . Confidence rises with population size; the uncertain labels (upper-left) are rare types with very few cells. Read any single-type claim against this map.

therefore plausibly an **undercount**, and the clean gate-pass overstates the stability of the off-target boundary. This is the highest-consequence Phase-1 task (§7, item 1), not a cosmetic detail.

- **Classifier over-training.** scANVI validation degraded past its optimum, and the class prior is uniform; rare-type labels are the least reliable (see confidence table).
- **Dish-vector is a proxy.** It is a mean-difference smell test, not the disentangled z_{iv} of the full method.
- **Gate C is GREEN but concentrated.** Verified explicitly (§5.4): four protocols clear 300 cells, so **GREEN** is genuine — yet 61.7% of the off-target mesenchyme sits in a single study and 72% in the top two. Pooled power is strong and four independent studies replicate the population, but broad protocol-stratified (Control 2) comparison is thin. A real, bounded caveat, not a disqualifier.

7 Next Steps: Phase 1

Phase 1 is the neural vertical, end-to-end, with the upgrades that convert geometry into mechanism. The ordering below reflects *scientific consequence*, not merely cost: two fixes must be front-loaded together, and one is the methodological centerpiece on which the eventual paper’s credibility rests.

Front-load immediately (in parallel)

1. **Source true raw UMI counts for the query — the single highest-consequence fix.** The per-dataset count matrices behind HNOCA, fed to genuine scArches surgery, change downstream validity more than anything else. The 37.7% **ambiguous_novel** is *not* cosmetic: more than a third of query cells land beyond the off-manifold gate, and an unknown fraction are real off-target cells that simply could not reach the reference manifold without query fine-tuning. Consequently the **true_offtarget** population (17.8%) is

plausibly **undercounted**, and the off-target definition carried into Aim 3 has a soft edge until this is resolved. Surgery should pull genuine off-target cells out of **ambiguous_novel** into **true_offtarget** and harden the gate. This outranks even the classifier hardening in scientific importance.

2. **Harden the classifier — cheap, necessary hygiene (~2 h).** Early-stopping / best-validation checkpoint and a non-uniform prior, fixing the rare-type label unreliability documented in §5.5. Recognize its scope, though: it mostly cleans up *rare* types, which are not the load-bearing populations (mesenchyme is abundant and sits at moderate entropy). It is hygiene, not the scientific crux — do it because it is cheap and upstream, not because it moves the headline.

The scientific centerpiece

3. **Maturation-matched Control 3 + contrastiveVI dish-stripping — what converts Gate B from encouraging to defensible.** The 0.906 cosine is the result to be most cautious about over-reading: some of that consistency may reflect shared normalization or technical structure rather than genuine in-vitro biology. Control 3 is the baseline that licenses the claim that the dish signature is *real biology* rather than a processing artifact — without it a reviewer can dismiss the central disentanglement claim; with it, the claim holds. contrastiveVI (background = primary on-target, target = organoid on-target) gives the identifiable, stable replacement for adversarial subspace separation; the adversary is demoted to a verification check. This is the methodological heart of the project.

Then build interpretation and scale

4. **In-silico levers (CellOracle) on the convergent population and in-vivo program anchoring** (signature scoring) — the analyses that turn a distance into an interpretable, actionable result.
5. **Build for scale:** geosketch subsampling and lean/backed I/O from the start, given the firm 251 GB container ceiling this pilot exposed.

8 Conclusion

The NullState pilot achieved its purpose. It produced a harmonized, integration-verified cross-atlas reference; numerically sound deep models; a behaving off-target definition; and a clean pass on all three decision gates (**GO** on consistency, **GREEN** on counts). The instrument is sound. The honest caveats — single system, no maturation control, projection-only mapping — are precisely the items Phase 1 is designed to resolve, and the pilot tells us exactly where to spend the next effort. The project is cleared to proceed.

Artifact manifest (local backup). Trained models (`scvi_model`, `ref_scanvi`, `query_scanvi` as `model.pt`); `integration.qc.json`; `label.confidence.csv`; training history (`training_progress.json`); run logs (`h100_run.log`, `h100_step34.log`); `pilot.log`. The raw `reference.h5ad` (10.8 GB) and `hnoca_annotated.h5ad` (18.7 GB) remain on the compute volume (not transferred due to size).